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DATA STRUCTURES AND ALGORITHMS – 3

Y25 - 2026-2027

Project Abstract Submission Form

Sec No- 1	Team No- 4	Project Title- Optimal Binary Search Tree

1. Abstract

The **Optimal Search Tree** is a software application designed to construct a Binary Search Tree using the search frequencies of different keys. In a normal Binary Search Tree, frequently searched keys may not always be placed in the best position, which can increase the overall search cost. This project aims to arrange the keys in an efficient way to reduce the average search cost.

The system takes a set of **distinct keys and their search frequencies** as input. It uses the **Dynamic Programming** technique to find the best possible arrangement of the keys. The program checks different possible root nodes and calculates the cost of each arrangement. It then selects the arrangement with the minimum expected search cost.

The application displays the resulting Optimal Binary Search Tree and shows its minimum search cost. This makes it easier for users to understand how the position of a key affects searching efficiency.

The main objective of this project is to demonstrate the practical use of Dynamic Programming in solving an optimization problem. Overall, the project provides a simple and easy-to-understand implementation of an Optimal Binary Search Tree and helps improve the efficiency of searching frequently accessed keys.

2. Team Details

Roll Number(s)	Name of the Student	Signature
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3.Problem Statement: In a Binary Search Tree, the position of a key affects the number of comparisons required to search for it. When different keys have different search frequencies, placing the keys in an unsuitable order can result in a higher average search cost.

The problem is to **construct an Optimal Binary Search Tree for a set of distinct keys and their search frequencies** such that the **expected search cost is minimized**. The project uses **Dynamic Programming** to consider different possible root nodes, calculate the search cost of each arrangement, and select the arrangement with the minimum cost.

The system should take the keys and their corresponding frequencies as input and produce the **Optimal Binary Search Tree and its minimum expected search cost**. This project demonstrates the use of Dynamic Programming to solve an optimization problem. This is consistent with the problem description and approach in your submitted DSA-3 abstract.

4.Frameworks / Platforms / Software Used

Programming Language: Java
Algorithm: Dynamic Programming
Data Structure: Binary Search Tree
IDE: IntelliJ IDEA / Eclipse / Visual Studio Code

Version Control: Git and GitHub